

Chapter 1 Groups and Homomorphisms

Exercise 1

Assume $H \neq \{1\}$. Let $g_1, g_2 \in G$. Suppose $g_1\vartheta = g_2\vartheta$. Then $(g_1\vartheta)(g_2\vartheta)^{-1} = 1$ and thus $(g_1g_2^{-1})\vartheta = 1$. Hence $g_1g_2^{-1} \in \text{Ker}(\vartheta)$. Since $\text{Ker}(\vartheta) \triangleleft G$, we have $\text{Ker}(\vartheta) = \{1\}$ or $\text{Ker}(\vartheta) = G$. If $\text{Ker}(\vartheta) = G$, then $g\vartheta = 1$ for all $g \in G$. Since ϑ is surjective, $H = \text{Im}(\vartheta) = \{g\vartheta \mid g \in G\} = \{1\}$, a contradiction. Therefore, $\text{Ker}(\vartheta) = \{1\}$. Hence, we have $g_1g_2^{-1} = 1$, which implies $g_1 = g_2$. So ϑ is injective and thus is bijective. $\square$